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(54) **INTERCONNECT STRUCTURE AND METHODS THEREOF**

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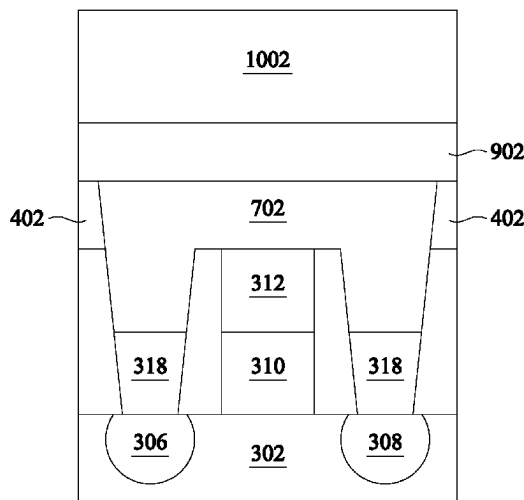
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(57) **ABSTRACT**

A method and structure for forming a local interconnect, without routing the local interconnect through an overlying metal layer. In various embodiments, a first dielectric layer is formed over a gate stack of at least one device and a second dielectric layer is formed over a contact metal layer of the at least one device. In various embodiments, a selective etching process is performed to remove the second dielectric layer and expose the contact metal layer, without substantial removal of the first dielectric layer. In some examples, a metal VIA layer is deposited over the at least one device. The metal VIA layer contacts the contact metal layer and provides a local interconnect structure. In some embodiments, a multi-level interconnect network overlying the local interconnect structure is formed.

20 Claims, 10 Drawing Sheets

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(58) Field of Classification Search CPC H01L 21/823431; H01L 23/535; H01L 29/41791; H01L 29/66553; H01L 29/66795; H01L 29/785; H01L 29/7851; H01L 21/76834; H01L 2221/1031; H10D 30/024; H10D 30/62; H10D 30/6211; H10D 30/6219; H10D 64/018; H10D 84/0158; H10D 84/038		2014/0312500 A1 * 10/2014 Zhu H01L 21/76895 257/773 2014/0319587 A1 10/2014 Chen et al. 2015/0085585 A1 3/2015 Horch et al. 2015/0170966 A1 6/2015 Tung 2016/0141291 A1 * 5/2016 Woo H01L 27/0207 438/294 2017/0162496 A1 6/2017 Chang et al.
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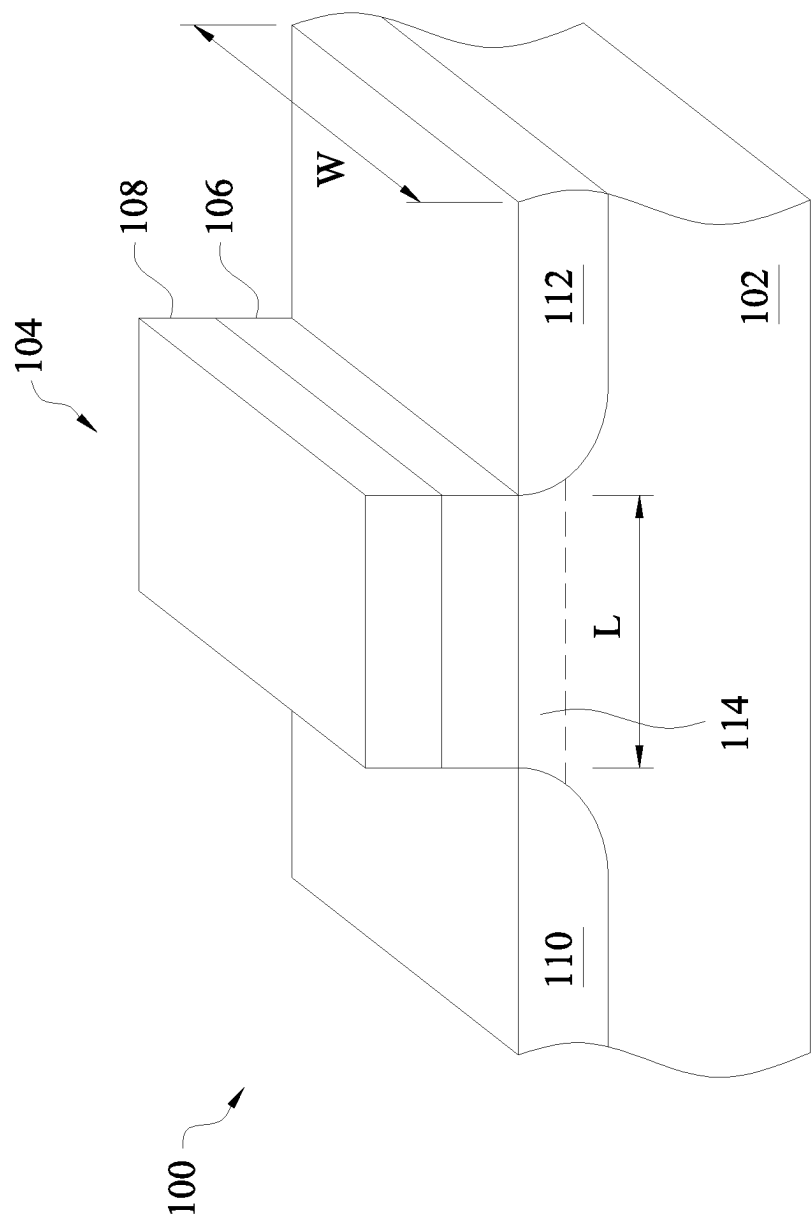


FIG. 1A

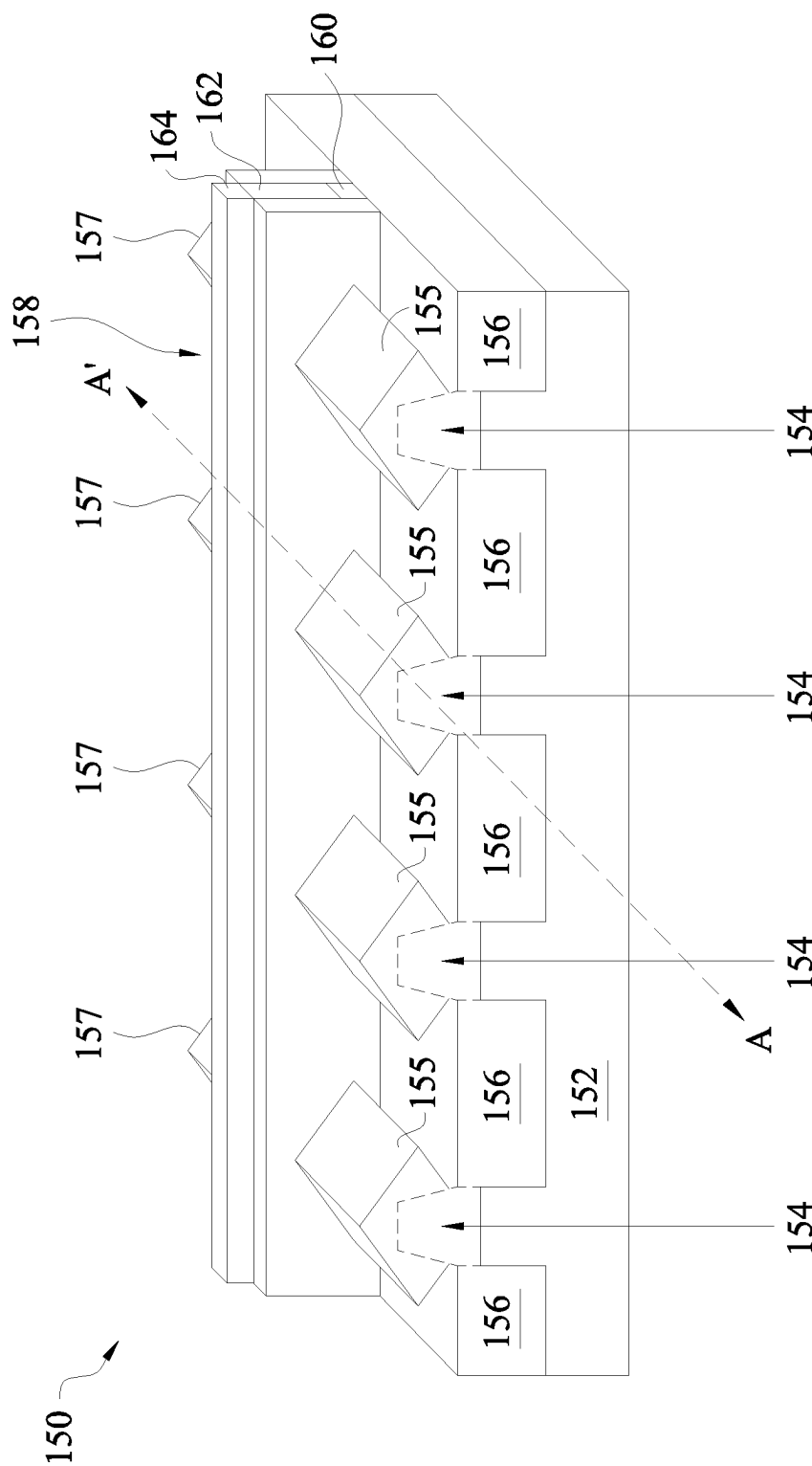


FIG. 1B

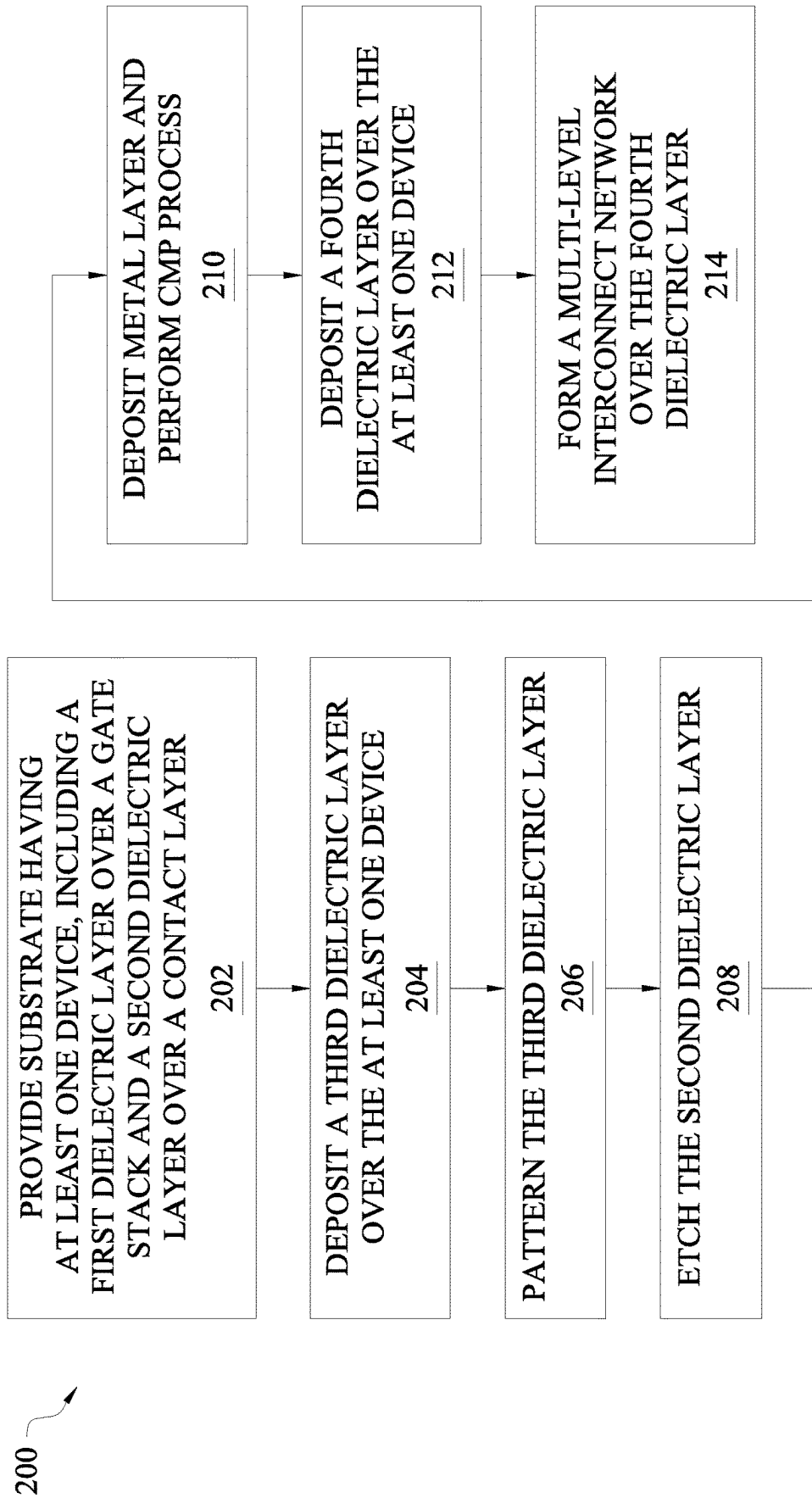


FIG. 2

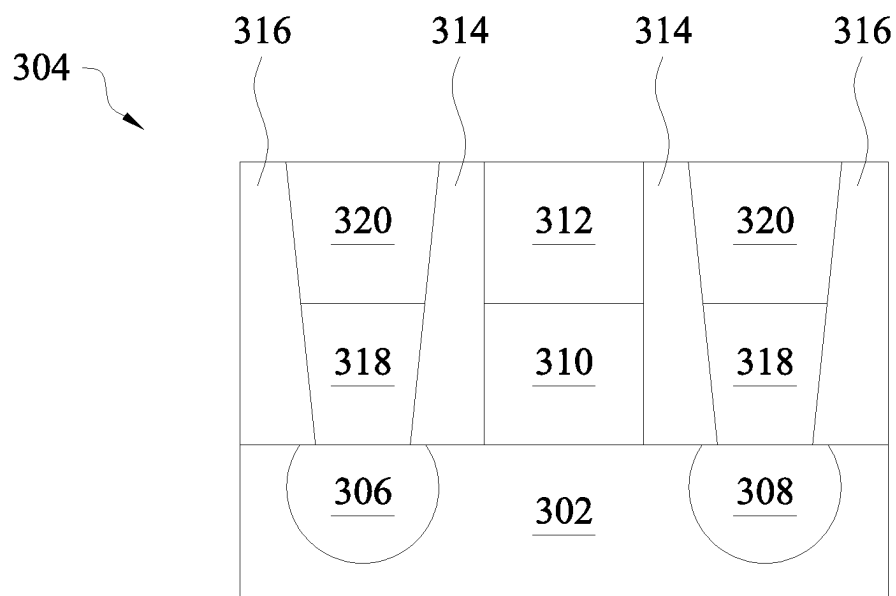


FIG. 3

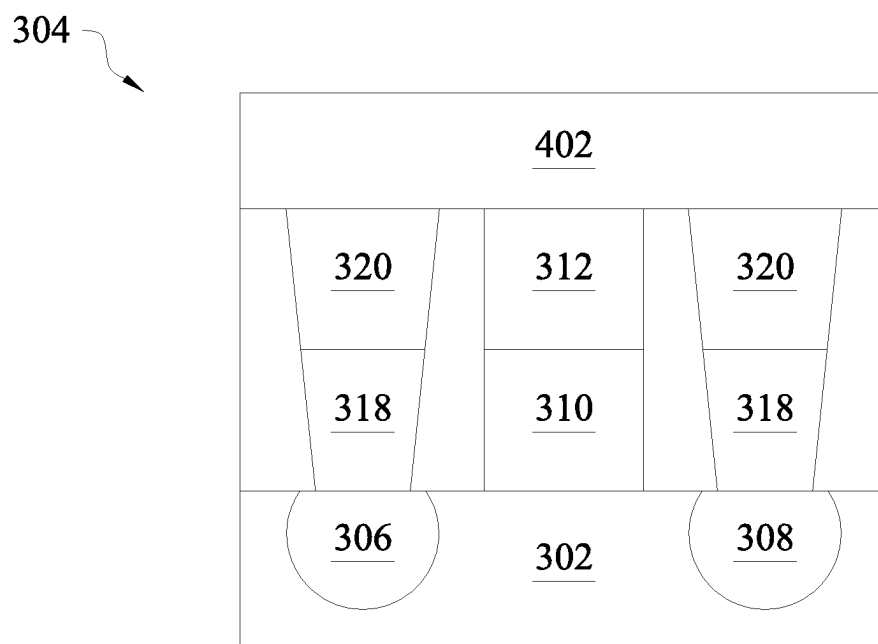


FIG. 4

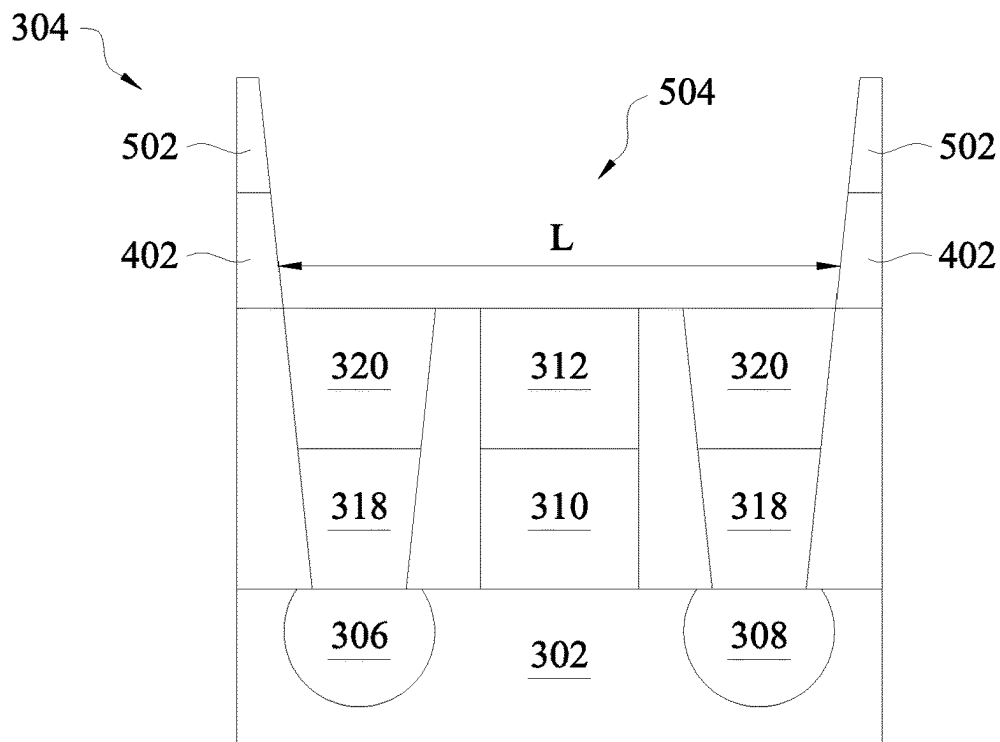


FIG. 5

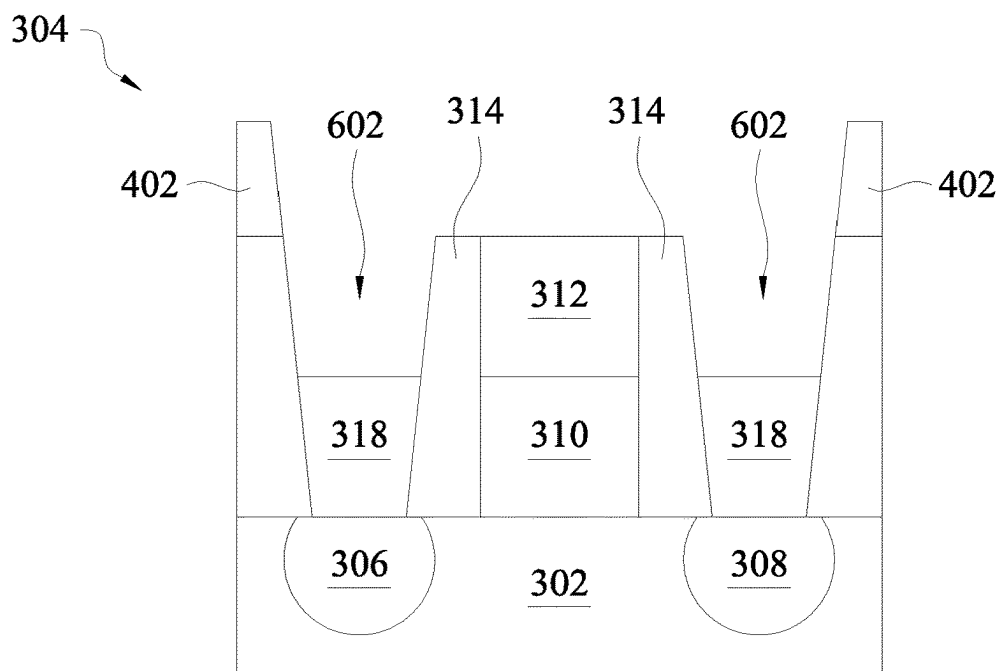


FIG. 6

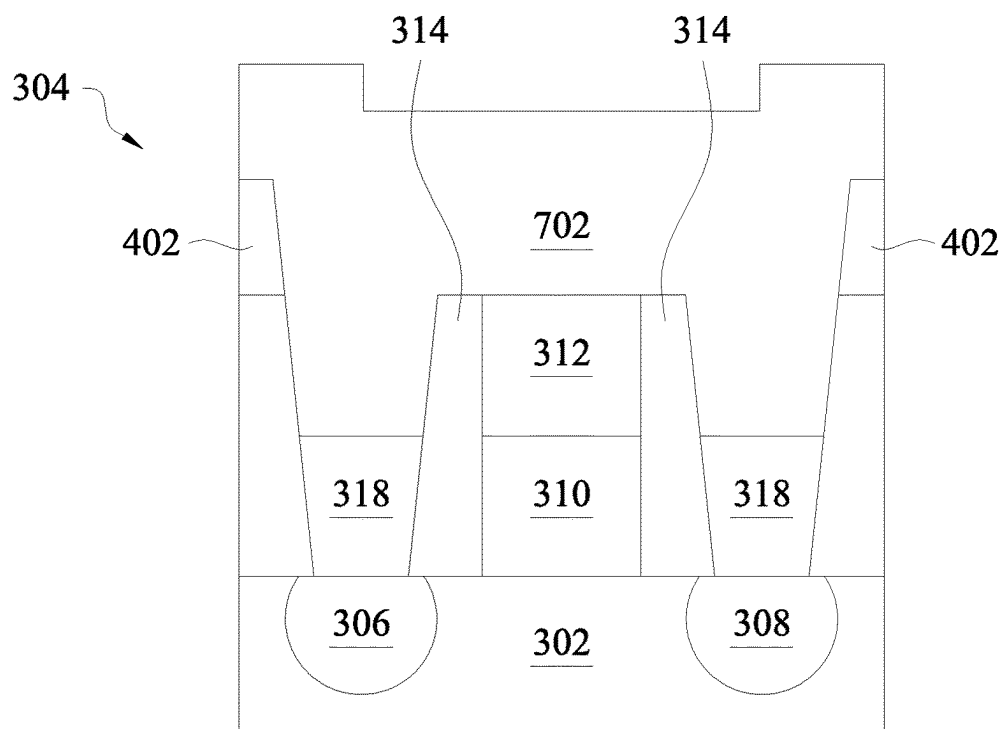


FIG. 7

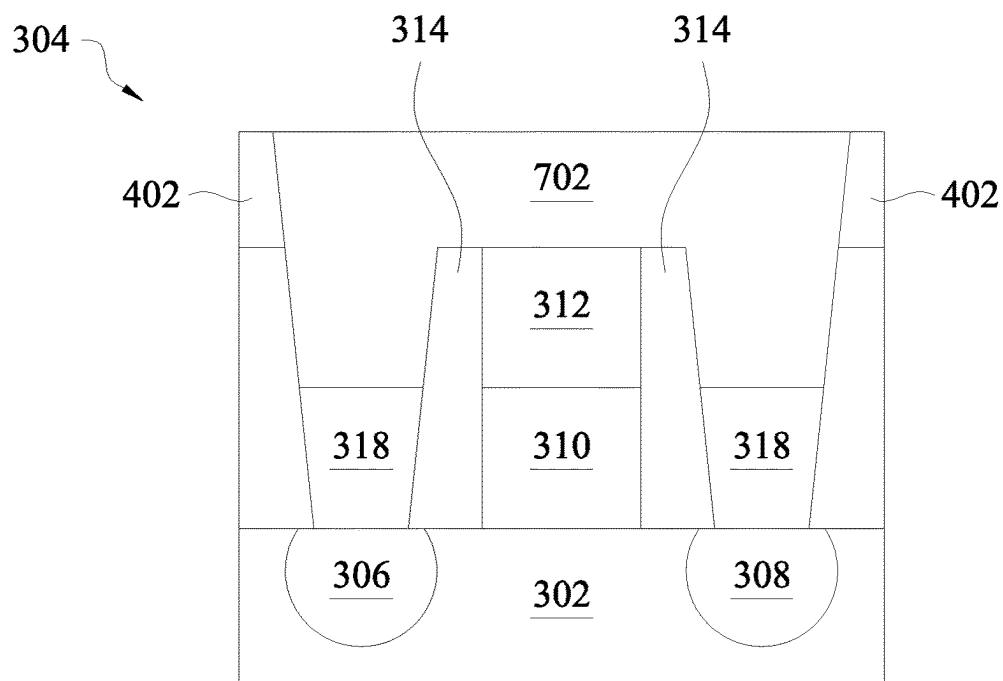


FIG. 8

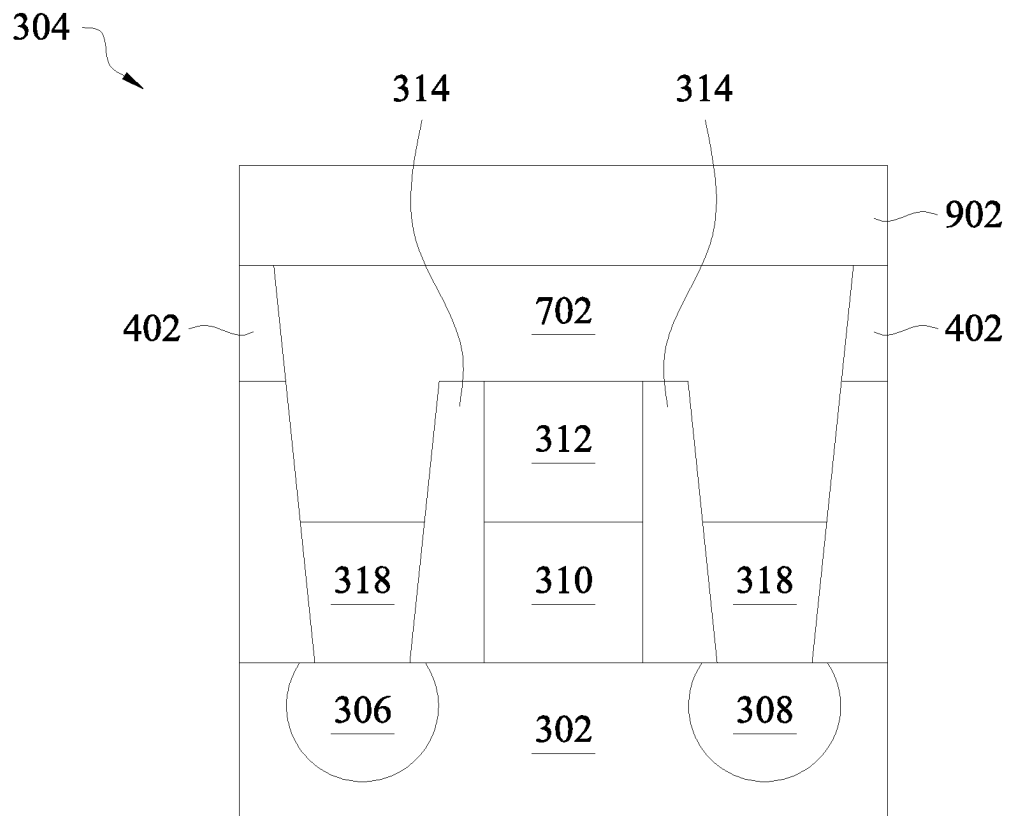


FIG. 9

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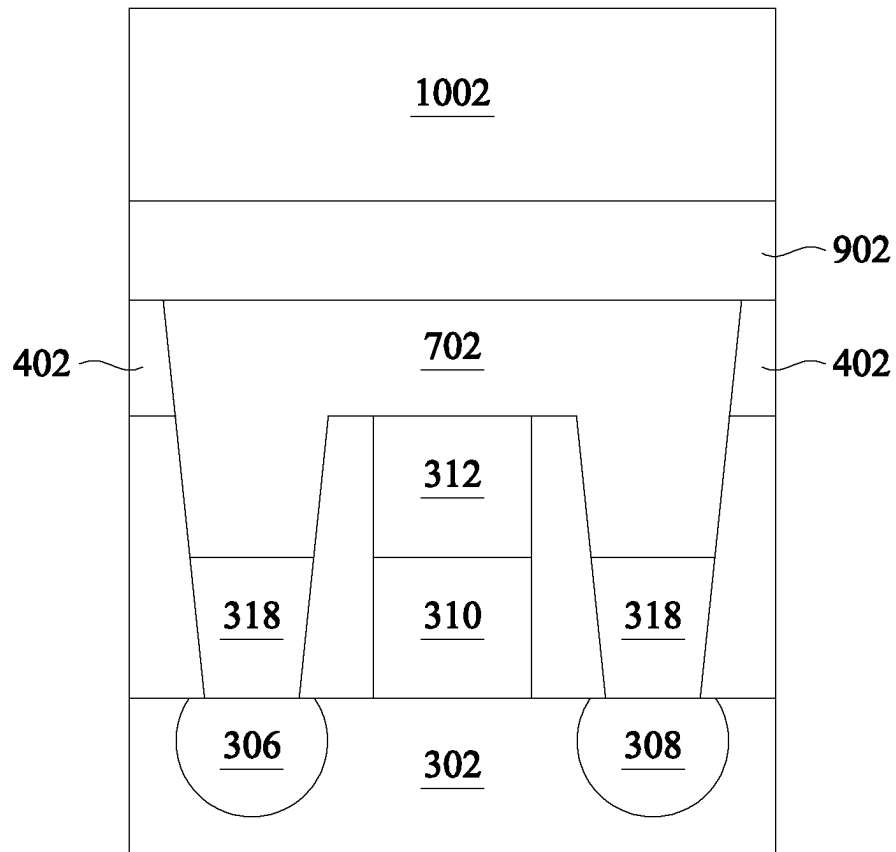



FIG. 10

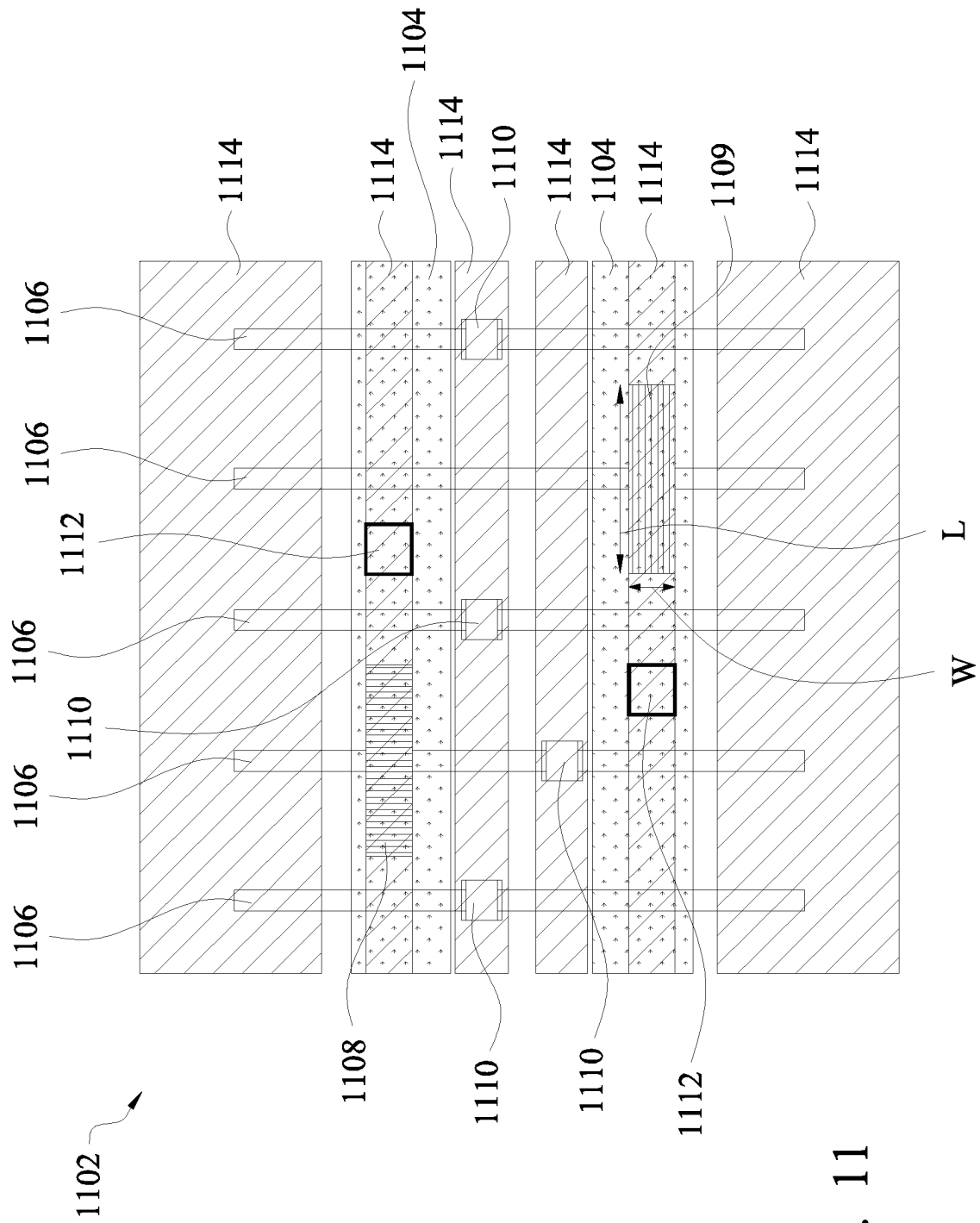


FIG. 11

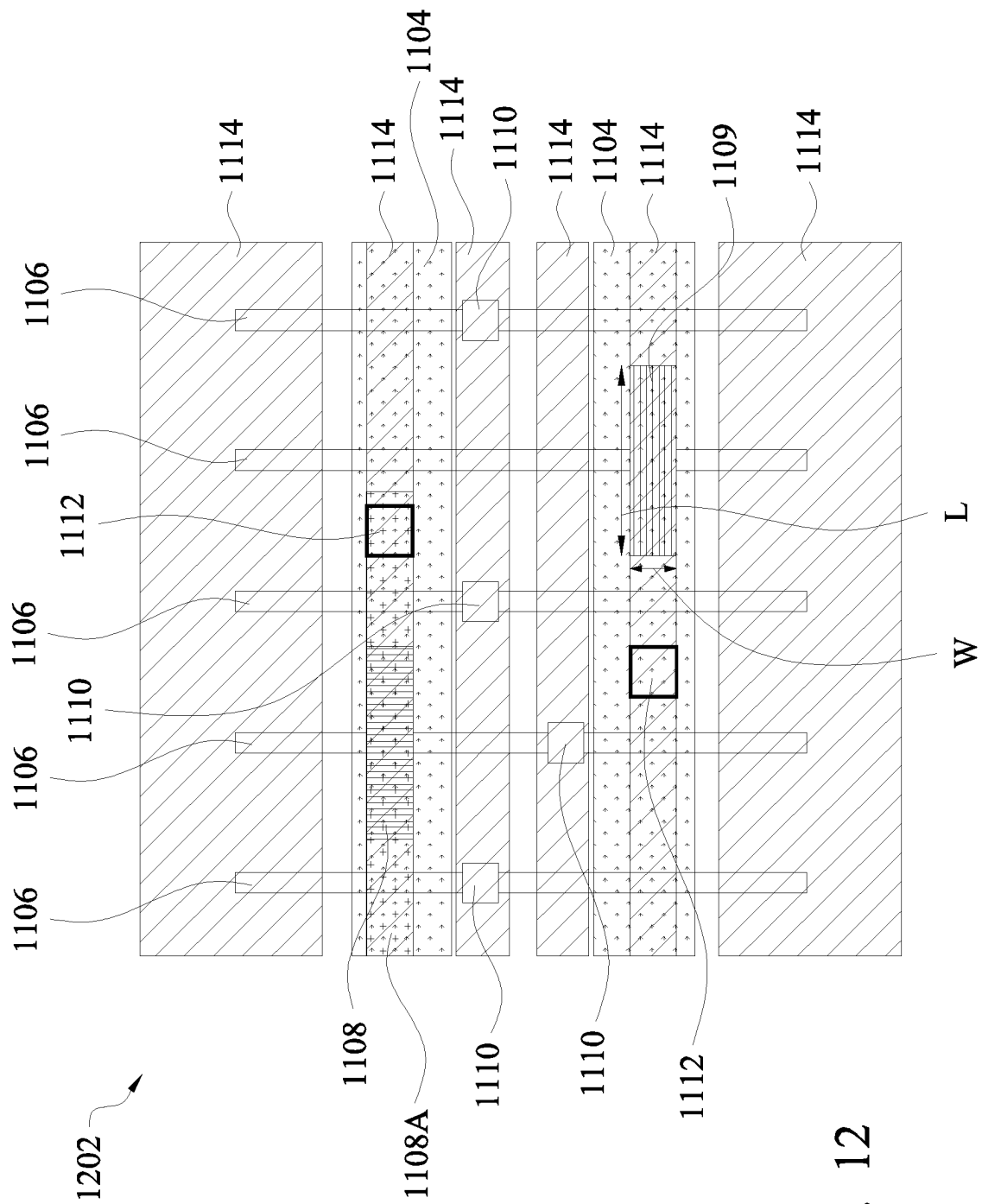


FIG. 12

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INTERCONNECT STRUCTURE AND METHODS THEREOF

PRIORITY DATA

The present application is a continuation application of U.S. patent application Ser. No. 16/047,884, filed Jul. 27, 2018, issuing as U.S. Pat. No. 11,222,842, which is a divisional application of U.S. patent application Ser. No. 15/253,311, filed Aug. 31, 2016, now U.S. Pat. No. 10,276,491, the entire disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND

The electronics industry has experienced an ever increasing demand for smaller and faster electronic devices which are simultaneously able to support a greater number of increasingly complex and sophisticated functions. Accordingly, there is a continuing trend in the semiconductor industry to manufacture low-cost, high-performance, and low-power integrated circuits (ICs). Thus far these goals have been achieved in large part by scaling down semiconductor IC dimensions (e.g., minimum feature size) and thereby improving production efficiency and lowering associated costs. However, such scaling has also introduced increased complexity to the semiconductor manufacturing process. Thus, the realization of continued advances in semiconductor ICs and devices calls for similar advances in semiconductor manufacturing processes and technology.

In particular, back-end-of-line (BEOL) fabrication processes have presented a particularly difficult challenge for advanced IC fabrication. BEOL fabrication includes interconnect structures, for example, made up of a multi-level network of metal wiring. Any of a plurality of IC circuits and/or devices may be connected by such interconnect structures. In various examples, however, interconnect performance degrades with dimensional scaling. For instance, resistance (R) increases as dimensions get smaller and capacitance (C) increases as a density of interconnects increases, both of which increase RC-delay. In some cases, different materials and/or processes for interconnect fabrication are being studied. In other examples, an increasing number of metal layers are being used as part of the multi-level interconnect network, to provide additional interconnect routing paths and potentially reduce R (e.g., by increasing interconnect dimensions) and/or reduce C (e.g., by reducing interconnect density). However, increasing the number of metal layers will invariably increase a total interconnect length, which can also degrade device performance. Moreover, the addition of metal layers will lead to an increase in cost (e.g., additional photomasks, design time, etc.).

Thus, existing techniques have not proved entirely satisfactory in all respects.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when they are read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1A is a cross-sectional view of an MOS transistor according to some embodiments;

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FIG. 1B is perspective view of an embodiment of a FinFET device according to one or more aspects of the present disclosure;

FIG. 2 is a flow chart of a method of forming a local interconnect, in accordance with some embodiments;

FIGS. 3-10 provide cross-sectional views of a device at intermediate stages of fabrication and processed in accordance with the method of FIG. 2; and

FIGS. 11 and 12 provide layout designs illustrating various aspects of embodiments of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

It is also noted that the present disclosure presents embodiments in the form of local interconnect structures which may be employed in any of a variety of device types. For example, embodiments of the present disclosure may be used to form local interconnect structures in planar bulk metal-oxide-semiconductor field-effect transistors (MOSFETs), multi-gate transistors (planar or vertical) such as FinFET devices, gate-all-around (GAA) devices, Omega-gate (Ω -gate) devices, or Pi-gate (Π -gate) devices, as well as strained-semiconductor devices, silicon-on-insulator (SOI) devices, partially-depleted SOI devices, fully-depleted SOI devices, or other devices as known in the art. In addition, embodiments disclosed herein may be employed in the formation of P-type and/or N-type devices. One of ordinary skill may recognize other embodiments of semiconductor devices that may benefit from aspects of the present disclosure.

With reference to the example of FIG. 1A, illustrated therein is an MOS transistor **100**, providing an example of merely one device type which may include embodiments of the present disclosure. It is understood that the exemplary transistor **100** is not meant to be limiting in any way, and those of skill in the art will recognize that embodiments of the present disclosure may be equally applicable to any of a variety of other device types, such as those described above. The transistor **100** is fabricated on a substrate **102** and

includes a gate stack **104**. The substrate **102** may be a semiconductor substrate such as a silicon substrate. The substrate **102** may include various layers, including conductive or insulating layers formed on the substrate **102**. The substrate **102** may include various doping configurations depending on design requirements as is known in the art. The substrate **102** may also include other semiconductors such as germanium, silicon carbide (SiC), silicon germanium (SiGe), or diamond. Alternatively, the substrate **102** may include a compound semiconductor and/or an alloy semiconductor. Further, in some embodiments, the substrate **102** may include an epitaxial layer (epi-layer), the substrate **102** may be strained for performance enhancement, the substrate **102** may include a silicon-on-insulator (SOI) structure, and/or the substrate **102** may have other suitable enhancement features.

The gate stack **104** includes a gate dielectric **106** and a gate electrode **108** disposed on the gate dielectric **130**. In some embodiments, the gate dielectric **106** may include an interfacial layer such as silicon oxide layer (SiO₂) or silicon oxynitride (SiON), where such interfacial layer may be formed by chemical oxidation, thermal oxidation, atomic layer deposition (ALD), chemical vapor deposition (CVD), and/or other suitable method. In some examples, the gate dielectric **106** includes a high-k dielectric layer such as hafnium oxide (HfO₂). Alternatively, the high-k dielectric layer may include other high-k dielectrics, such as TiO₂, HfZrO, Ta₂O₃, HfSiO₄, ZrO₂, ZrSiO₂, LaO, AlO, ZrO, TiO, Ta₂O₅, Y₂O₃, SrTiO₃ (STO), BaTiO₃ (BTO), BaZrO, HfZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfTiO, (Ba,Sr)TiO₃ (BST), Al₂O₃, Si₃N₄, oxynitrides (SiON), combinations thereof, or other suitable material. High-K gate dielectrics, as used and described herein, include dielectric materials having a high dielectric constant, for example, greater than that of thermal silicon oxide (~3.9). In still other embodiments, the gate dielectric **106** may include silicon dioxide or other suitable dielectric. The gate dielectric **106** may be formed by ALD, physical vapor deposition (PVD), CVD, oxidation, and/or other suitable methods. In some embodiments, the gate electrode **108** may be deposited as part of a gate first or gate last (e.g., replacement gate) process. In various embodiments, the gate electrode **108** includes a conductive layer such as W, Ti, TiN, TiAl, TiAlN, Ta, TaN, WN, Re, Ir, Ru, Mo, Al, Cu, Co, CoSi, Ni, NiSi, combinations thereof, and/or other suitable compositions. In some examples, the gate electrode **108** may include a first metal material for an N-type transistor and a second metal material for a P-type transistor. Thus, the transistor **100** may include a dual work-function metal gate configuration. For example, the first metal material (e.g., for N-type devices) may include metals having a work function substantially aligned with a work function of the substrate conduction band, or at least substantially aligned with a work function of the conduction band of a channel region **114** of the transistor **100**. Similarly, the second metal material (e.g., for P-type devices) may include metals having a work function substantially aligned with a work function of the substrate valence band, or at least substantially aligned with a work function of the valence band of the channel region **114** of the transistor **100**. Thus, the gate electrode **104** may provide a gate electrode for the transistor **100**, including both N-type and P-type devices. In some embodiments, the gate electrode **108** may alternately or additionally include a polysilicon layer. In various examples, the gate electrode **108** may be formed using PVD, CVD, electron beam (e-beam) evaporation, and/or other suitable process. In some embodiments, sidewall spacers are formed on sidewalls of the gate stack

104. Such sidewall spacers may include a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, or combinations thereof.

The transistor **100** further includes a source region **110** and a drain region **112** each formed within the semiconductor substrate **102**, adjacent to and on either side of the gate stack **104**. In some embodiments, the source and drain regions **110**, **112** include diffused source/drain regions, ion implanted source/drain regions, epitaxially grown regions, or a combination thereof. The channel region **114** of the transistor **100** is defined as the region between the source and drain regions **110**, **112** under the gate dielectric **106**, and within the semiconductor substrate **102**. The channel region **114** has an associated channel length "L" and an associated channel width "W". When a bias voltage greater than a threshold voltage (V_t) (i.e., turn-on voltage) for the transistor **100** is applied to the gate electrode **108** along with a concurrently applied bias voltage between the source and drain regions **110**, **112**, an electric current (e.g., a transistor drive current) flows between the source and drain regions **110**, **112** through the channel region **114**. The amount of drive current developed for a given bias voltage (e.g., applied to the gate electrode **108** or between the source and drain regions **110**, **112**) is a function of, among others, the mobility of the material used to form the channel region **114**. In some examples, the channel region **114** includes silicon (Si) and/or a high-mobility material such as germanium, which may be epitaxially grown, as well as any of the plurality of compound semiconductors or alloy semiconductors as known in the art. High-mobility materials include those materials with electron and/or hole mobility greater than silicon (Si), which has an intrinsic electron mobility at room temperature (300 K) of around 1350 cm²/V-s and a hole mobility of around 480 cm²/V-s.

Referring to FIG. 1B, illustrated therein is a FinFET device **150**, providing an example of an alternative device type which may include embodiments of the present disclosure. By way of example, the FinFET device **150** includes one or more fin-based, multi-gate field-effect transistors (FETs). The FinFET device **150** includes a substrate **152**, at least one fin element **154** extending from the substrate **152**, isolation regions **156**, and a gate structure **158** disposed on and around the fin-element **154**. The substrate **152** may be a semiconductor substrate such as a silicon substrate. In various embodiments, the substrate **152** may be substantially the same as the substrate **102**, as described above.

The fin-element **154**, like the substrate **152**, may include one or more epitaxially-grown layers, and may comprise silicon or another elementary semiconductor, such as germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, InGaAs, GaInP, and/or GaInAsP; or combinations thereof. The fins **154** may be fabricated using suitable processes including photolithography and etch processes. The photolithography process may include forming a photoresist layer (resist) overlying the substrate (e.g., on a silicon layer), exposing the resist to a pattern, performing post-exposure bake processes, and developing the resist to form a masking element including the resist. In some embodiments, patterning the resist to form the masking element may be performed using an electron beam (e-beam) lithography process. The masking element may then be used to protect regions of the substrate while an etch process forms recesses into the silicon layer, thereby leaving an extending fin **154**. The recesses may be etched using a dry etch (e.g., chemical

oxide removal), a wet etch, and/or other suitable processes. Numerous other embodiments of methods to form the fins **154** on the substrate **152** may also be used.

Each of the plurality of fins **154** also include a source region **155** and a drain region **157** where the source/drain regions **155**, **157** are formed in, on, and/or surrounding the fin **154**. The source/drain regions **155**, **157** may be epitaxially grown over the fins **154**. In addition, a channel region of a transistor is disposed within the fin **154**, underlying the gate structure **158**, along a plane substantially parallel to a plane defined by section AA' of FIG. 1B. In some examples, the channel region of the fin includes a high-mobility material, as described above.

The isolation regions **156** may be shallow trench isolation (STI) features. Alternatively, a field oxide, a LOCOS feature, and/or other suitable isolation features may be implemented on and/or within the substrate **152**. The isolation regions **156** may be composed of silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric, combinations thereof, and/or other suitable material known in the art. In an embodiment, the isolation structures are STI features and are formed by etching trenches in the substrate **152**. The trenches may then be filled with isolating material, followed by a chemical mechanical polishing (CMP) process. However, other embodiments are possible. In some embodiments, the isolation regions **156** may include a multi-layer structure, for example, having one or more liner layers.

The gate structure **158** includes a gate stack having an interfacial layer **160** formed over the channel region of the fin **154**, a gate dielectric layer **162** formed over the interfacial layer **160**, and a metal layer **164** formed over the gate dielectric layer **162**. In various embodiments, the interfacial layer **160** is substantially the same as the interfacial layer described as part of the gate dielectric **106**. In some embodiments, the gate dielectric layer **162** is substantially the same as the gate dielectric **106** and may include high-k dielectrics similar to that used for the gate dielectric **106**. Similarly, in various embodiments, the metal layer **164** is substantially the same as the gate electrode **108**, described above. In some embodiments, sidewall spacers are formed on sidewalls of the gate structure **158**. The sidewall spacers may include a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, or combinations thereof.

As discussed above, each of the transistor **100** and FinFET device **150** may include one or more local interconnect structures, embodiments of which are described in more detail below. As used herein, the term "local interconnect" is used to describe the lowest level of metal interconnects and are differentiated from intermediate and/or global interconnects. Local interconnects span relatively short distances and are sometimes used, for example, to electrically connect a source, drain, and/or gate of a given device, or those of nearby devices. Additionally, local interconnects may be used to facilitate a vertical connection of one or more devices to an overlying metallization layer (e.g., to an intermediate interconnect layer), for example, through one or more vias. Interconnects (e.g., including local, intermediate, or global interconnects), in general, are formed as part of back-end-of-line (BEOL) fabrication processes and include a multi-level network of metal wiring. Moreover, any of a plurality of IC circuits and/or devices (e.g., such as the transistor **100** or FinFET **150**) may be connected by such interconnects.

With the aggressive scaling and ever-increasing complexity of advanced IC devices and circuits, interconnect design and performance has proved to be a difficult challenge. For

example, with respect to dimensional scaling, resistance (R) (e.g., of a given interconnect) increases as dimensions get smaller and capacitance (C) (e.g., of the given interconnect) increases as a density of interconnects increases, both of which increase RC-delay. In some examples, additional metal layers have been used as part of the multi-level interconnect network, for example, to provide additional interconnect routing paths and potentially reduce R (e.g., by increasing interconnect dimensions) and/or reduce C (e.g., by reducing interconnect density). For instance, in some cases, local interconnects may be routed through an overlying metal interconnect layer. As merely one example, consider a scenario in which one would like to connect a source and a drain of a given device (e.g., such as the transistor **100** or FinFET **150**), for example using a local interconnect. In at least some existing methods, the local interconnect routing between the source and drain may go through an overlying metal layer, as discussed above. In short, routing a local interconnect through an overlying metal layer is inefficient, costly, and can degrade device and/or circuit performance. This is demonstrated at least by the fact that increasing the number of metal layers (e.g., to provide the additional interconnect routing paths), will invariably increase a total interconnect length, which can also degrade device performance. Moreover, the addition of metal layers will lead to an increase in cost, for example, due to the need for additional photomasks and increased design time, among other factors. Thus, at least some existing methods of forming local interconnect have not been entirely satisfactory in all respects.

Embodiments of the present disclosure offer advantages over the existing art, though it is understood that other embodiments may offer different advantages, not all advantages are necessarily discussed herein, and no particular advantage is required for all embodiments. For example, embodiments discussed herein include methods and structures directed to a local interconnect fabrication process that provides a local interconnect without routing the local interconnect through an overlying metal layer. In particular, and in at least some embodiments, a local interconnect fabrication process is provided which provides the local interconnect routing through a VIA layer, without having to utilize interconnect layers of the multi-level network of metal wiring. As a result, and in various embodiments, the number of metal layers used in the multi-level interconnect network may be reduced. In some examples, the number of metal layers may be reduced at least by one. As a result of providing the local interconnect routing through the VIA layer, rather than routing through layers of the multi-level interconnect network, the multi-level interconnect network, overlying the local interconnect, will have improved routing efficiency. Moreover, and in comparison to at least some existing solutions, embodiments disclosed herein reduce cost (e.g., by reduction of a metal layer) and provide for improved device and/or circuit performance (e.g., by way of a decreased interconnect length). It is also noted that the embodiments disclosed herein may be equally applied to both single-height and double-height cell architectures. Additional details of embodiments of the present disclosure are provided below, and additional benefits and/or other advantages will become apparent to those skilled in the art having benefit of the present disclosure.

Referring now to FIG. 2, illustrated is a method **200** of forming a local interconnect without routing the local interconnect through an overlying metal layer, in accordance with some embodiments. The method **200** is described below in more detail with reference to FIGS. 3-10. The

method **200** may be implemented on a single-gate planar device, such as the exemplary transistor **100** described above with reference to FIG. 1A, as well as on a multi-gate device, such as the FinFET device **150** described above with reference to FIG. 1B. Thus, one or more aspects discussed above with reference to the transistor **100** and/or the FinFET **150** may also apply to the method **200**. To be sure, in various embodiments, the method **200** may be implemented on other devices such as gate-all-around (GAA) devices, Omega-gate (Ω -gate) devices, or Pi-gate (Π -gate) devices, as well as strained-semiconductor devices, silicon-on-insulator (SOI) devices, partially-depleted SOI (PD-SOI) devices, fully-depleted SOI (FD-SOI) devices, or other devices as known in the art.

It is understood that parts of the method **200** and/or any of the exemplary transistor devices discussed with reference to the method **200** may be fabricated by a well-known complementary metal-oxide-semiconductor (CMOS) technology process flow, and thus some processes are only briefly described herein. Further, it is understood that any exemplary transistor devices discussed herein may include various other devices and features, such as additional transistors, bipolar junction transistors, resistors, capacitors, diodes, fuses, etc., but are simplified for a better understanding of the inventive concepts of the present disclosure. Further, in some embodiments, the exemplary transistor device(s) disclosed herein may include a plurality of semiconductor devices (e.g., transistors), which may be interconnected. In addition, in some embodiments, various aspects of the present disclosure may be applicable to either one of a gate-last process or a gate-first process.

In addition, in some embodiments, the exemplary transistor devices illustrated herein may include a depiction of a device at an intermediate stage of processing, as may be fabricated during processing of an integrated circuit, or portion thereof, that may comprise static random access memory (SRAM) and/or other logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as P-channel field-effect transistors (PFETs), N-channel FETs (NFETs), metal-oxide-semiconductor field-effect transistors (MOSFETs), complementary metal-oxide-semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, other memory cells, and/or combinations thereof.

The method **200** begins at block **202** where a substrate having at least one device, including a first dielectric layer over a gate stack and a second dielectric layer over a contact layer, is provided. With reference to FIG. 3, and in an embodiment of block **202**, a substrate **302** including a device **304** is provided. In some embodiments, the substrate **302** may be substantially the same as either of the substrates **102**, **152**, described above. It will be understood that the device **304** is merely illustrative, and is provided for clarity of discussion regarding subsequent formation of the local interconnect. For example, in some cases, the device **304** may include a planar device, such as the transistor **100**. Alternatively, in some examples, the device **304** may include a multi-gate device, such as the FinFET **150**. Moreover, in some cases, the device **304** may include a GAA device, an Ω -gate device, a Π -gate device, a strained-semiconductor device, an SOI device, a PD-SOI device, a FD-SOI device, or other device as known in the art. In some embodiments, the device **304** includes a source **306**, a drain **308**, and a gate stack **310**. The device **304** may also include a channel region between the source **306** and the drain **308**, under the gate stack **310**, and within the substrate **302**. In various embodiments, the gate stack **310** may include an interfacial layer

formed over the channel region, a gate dielectric layer formed over the interfacial layer, and a metal layer formed over the gate dielectric layer. In some embodiments, each of the interfacial layer, the dielectric layer, and the metal layer of the gate stack **310** may be substantially the same as those described above with respect to the transistor **100** and the FinFET **150**.

As shown in FIG. 3, the device **304** may further include a first dielectric layer **312** over the gate stack **310**. In some embodiments, the first dielectric layer **312** includes SiO_x , SiN , SiO_xN_y , SiC_xN_y , $\text{SiO}_x\text{C}_y\text{N}_z$, AlO_x , AlO_xN_y , AlN , HfO , ZrO , HfZrO , CN , poly-Si, combinations thereof, or other suitable dielectric material. The first dielectric layer **312** may be formed by ALD, PVD, CVD, oxidation, and/or other suitable methods. In some examples, the first dielectric layer **312** may have a thickness in a range of about 5-25 nanometers. As discussed in more detail below, the first dielectric layer **312** separates the gate stack **310** from the subsequently formed local interconnect. In various embodiments, sidewall spacers **314** are disposed on the sidewalls of the gate stack **310** and the first dielectric layer **312**. In some embodiments, the sidewall spacers **314** include SiO_x , SiN , SiO_xN_y , SiC_xN_y , $\text{SiO}_x\text{C}_y\text{N}_z$, AlO_x , AlO_xN_y , AlN , HfO , ZrO , HfZrO , CN , poly-Si, combinations thereof, or other suitable dielectric material. In some embodiments, the sidewall spacers **314** include multiple layers, such as main spacer walls, liner layers, and the like. By way of example, the sidewall spacers **314** may be formed by depositing a dielectric material over the device **304** and anisotropically etching back the dielectric material. In some embodiments, the etch-back process (e.g., for spacer formation) may include a multiple-step etching process to improve etch selectivity and provide over-etch control.

Additionally, in some embodiments, an inter-layer dielectric (ILD) layer **316** is formed over the device **304**. By way of example, the ILD layer **316** may include materials such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. The ILD layer **316** may be deposited by a subatmospheric CVD (SACVD) process, a flowable CVD process, or other suitable deposition technique. In some cases, a contact etch stop layer (CESL) may be deposited prior to deposition of the ILD layer **316**.

In various embodiments, contact openings are then formed (e.g., within the ILD layer **316**, and in some cases, a portion of the sidewall spacers **314**). For example, source/drain contact openings may be formed to provide access to the source and drain **306**, **308**. By way of example, the source/drain contact openings may be formed by a suitable combination of lithographic patterning and etching (e.g., wet or dry etching) processes. Thereafter, a source/drain contact metal is formed within the source/drain contact openings, thereby providing an electrical connection to the source and drain **306**, **308**. In particular, as shown in the example of FIG. 3, a source/drain contact metal **318** is formed, for example by a suitable combination of layer deposition (e.g., PVD, ALD, CVD), lithographic patterning and etching (e.g., wet or dry etching) processes. In some embodiments, the source/drain contact metal **318** includes Ti, W, Co, Cu, Al, Mo, MoW, W, TiN, TaN, WN, silicides, combinations thereof, or other suitable conductive material. In some examples, the device **304** may further include a second dielectric layer **320** over the source/drain contact metal **318**. In some embodiments, the second dielectric layer **320** includes SiO_x , SiN , SiO_xN_y , SiC_xN_y , $\text{SiO}_x\text{C}_y\text{N}_z$, AlO_x ,

AlO_xN_y , AlN , HfO , ZrO , HfZrO , CN , poly-Si, combinations thereof, or other suitable dielectric material. The second dielectric layer 320 may be formed by ALD, PVD, CVD, oxidation, and/or other suitable methods. In some examples, the second dielectric layer 320 may have a thickness in a range of about 5-25 nanometers. In some embodiments, a chemical mechanical planarization (CMP) process may be performed to remove excess material and planarize the top surface of the device 304.

It is noted that conventionally, a conductive layer may be formed over the source/drain contact metal 318 immediately after formation of the source/drain contact metal 318. Initial formation of the second dielectric layer 320 over the source/drain contact metal 318, and later removal as discussed in more detail below, enables subsequent formation of the local interconnect, in accordance with embodiments of the present disclosure.

The method 200 proceeds to block 204 where a third dielectric layer is deposited over the at least one device. With reference to FIG. 4, and in an embodiment of block 204, a third dielectric layer 402 is formed over the device 304. In some embodiments, the third dielectric layer 402 includes SiO_x , SiN , SiO_xN_y , SiC_xN_y , $\text{SiO}_x\text{C}_y\text{N}_z$, AlO_x , AlO_xN_y , AlN , HfO , ZrO , HfZrO , CN , poly-Si, combinations thereof, or other suitable dielectric material. The third dielectric layer 402 may be formed by ALD, PVD, CVD, oxidation, and/or other suitable methods. In some examples, the third dielectric layer 402 may have a thickness in a range of about 5-25 nanometers.

The method 200 then proceeds to block 206 where the third dielectric layer is patterned. With reference to FIGS. 4 and 5, and in an embodiment of block 206, the third dielectric layer 402 is patterned by a photolithography and etching process. As part of the patterning process, and in various embodiments, a photoresist layer 502 may first be deposited over the third dielectric layer 402 (e.g., by spin-on coating). In some examples, a hard mask layer may optionally be deposited over the third dielectric layer 402, with the photoresist layer 502 then formed over the hard mask layer. In embodiments including a hard mask layer, the hard mask layer may include a suitable dielectric material, such as silicon nitride, silicon oxynitride or silicon carbide, or other appropriate material. The hard mask layer may be formed by ALD, PVD, CVD, and/or other suitable methods. In some examples, after forming the photoresist layer 502, additional photolithography steps may include soft baking, mask aligning, exposure, post-exposure baking, developing, rinsing, drying (e.g., spin-drying and/or hard baking), other suitable lithography techniques, and/or combinations thereof. As a result of the photolithography process, a pattern is formed within the photoresist layer 502, where such a pattern may then be used as a mask to etch the underlying third dielectric layer 402. In particular, and as shown in FIG. 5, an etching process is performed to etch the third dielectric layer 402, thereby transferring the pattern of the photoresist layer 502 to the third dielectric layer 402 and forming a slot 504. In various embodiments, the etching process may include a dry etching process (e.g., RIE or ICP etching), a wet etching process, or a combination thereof. It is also noted that the etching process employed may include a selective etching process, such as a selective wet or selective dry etching process, that provides for removal of desired portions of the third dielectric layer 402, without substantial removal of other layers that may be exposed to the etching process (e.g., the first dielectric layer 312, the sidewall spacers 314, and/or the second dielectric layer 320). In addition, the slot 504 may be patterned in a variety of sizes, in accordance with a

particular application, technology, or other process requirement. By way of example, in some cases, the slot 504 may be patterned to have a slot length 'L' equal to about 40-80 nanometers and a slot width 'W' equal to about 5-20 nanometers. Other slot geometries are likewise envisioned as falling within the scope of this disclosure, as discussed in more detail below with reference to FIGS. 11 and 12. In various embodiments, the patterning of the slot will determine, at least in part, a size (e.g., length and width) of the subsequently formed local interconnect, as discussed below.

The method 200 then proceeds to block 208 where the second dielectric layer is etched. Additionally, in various examples, the photoresist layer 502 is removed after patterning the third dielectric layer 402 (block 206) and prior to etching the second dielectric layer 320 (block 208). With reference to FIGS. 5 and 6, and in an embodiment of block 208, the second dielectric layer 320 is etched to form openings 602 and thereby provide access to the source/drain contact metal 318. In particular, in various embodiments, the second dielectric layer 320 may be etched using a selective etching process, such as a selective wet or selective dry etching process, that provides for removal of the second dielectric layer 320, without substantial removal of other layers that may be exposed to the etching process (e.g., the first dielectric layer 312, the sidewall spacers 314, and/or the third dielectric layer 402).

In some embodiments, the second dielectric layer 320 and the third dielectric layer 402 may include different materials that are not equally etched by a given wet or dry etching process. In such cases, the method 200 may proceed as described above, with distinct etching processes for each of the second dielectric layer 320 and the third dielectric layer 402. However, in some cases, the second dielectric layer 320 and the third dielectric layer 402 may include the same material, or materials that are substantially equally etched by a given wet or dry etching process, such that a single etching process may be used to etch the third dielectric layer 402 and the underlying second dielectric layer 320. Regardless of whether a single etching process or distinct etching processes are used to etch the second dielectric layer 320 and the third dielectric layer 402, the process includes a selective etching process that does not substantially remove at least the first dielectric layer 312 and the sidewall spacers 314.

The method 200 then proceeds to block 210 where a metal layer is deposited and a CMP process is performed. Referring to FIG. 7, and in an embodiment of block 210, a metal layer 702 is deposited by ALD, PVD, CVD, and/or other suitable method. By way of example, the metal layer 702 includes a blanket layer of metal deposited over the device 304. In some embodiments, the metal layer 702 includes Ti, W, Co, Cu, Al, Mo, MoW, W, TiN, TaN, WN, silicides, combinations thereof, or other suitable conductive material. As shown in FIG. 7, the deposited metal layer 702 fills the slot 504 and the openings 602 and contacts the source/drain contact metal 318 over each of the source 306 and the drain 308, thereby electrically connecting the source 306 and the drain 308 of the device 304 and thus providing a local interconnect. In addition, the first dielectric layer 312 separates, and in some cases electrically isolates, the gate stack 310 from the metal layer 702 that provides the local interconnect. With reference to FIG. 8, and also in an embodiment of block 210, a CMP process is performed to remove excess material (e.g., excess material of the metal layer 702) and planarize the top surface of the device 304. In various examples, the metal layer 702 includes a layer conventionally used as a VIA (e.g., that would conventionally connect the source/drain contact metal 318 to an overlying metal

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layer that is part of a conventional multi-level interconnect network). However, in accordance with embodiments of the present disclosure, rather than routing through such an overlying metal layer, the local interconnect is provided through a VIA layer (e.g., the metal layer 702), without having to utilize an overlying metal layer. Thus, routing efficiency is improved, a number of required metal layers is reduced (e.g., at least by one metal layer), cost is reduced, and device performance is improved.

The method 200 proceeds to block 212 where a fourth dielectric layer is deposited over the at least one device. With reference to FIG. 9, and in an embodiment of block 212, a fourth dielectric layer 902 is formed over the device 304, including over the metal layer 702 that provides the local interconnect. In some embodiments, the fourth dielectric layer 902 includes SiO_x , SiN , SiO_xN_y , SiC_xN_y , $\text{SiO}_x\text{C}_y\text{N}_z$, AlO_x , AlO_xN_y , AlN , HfO , ZrO , HfZrO , CN , poly-Si, combinations thereof, or other suitable dielectric material. The fourth dielectric layer 902 may be formed by ALD, PVD, CVD, oxidation, and/or other suitable methods. In some examples, the fourth dielectric layer 902 may have a thickness in a range of about 5-25 nanometers.

The method 200 proceeds to block 214 where a multi-level interconnect network is formed over the fourth dielectric layer. With reference to FIGS. 9 and 10, and in an embodiment of block 214, a multi-level interconnect network 1002 is formed over the fourth dielectric layer 902. In some cases, the multi-level interconnect network 1002 includes intermediate and global interconnects, while local interconnects are provided in accordance with embodiments of the present disclosure (e.g., such as described for the local interconnect provided by the metal layer 702). In some embodiments, the multi-level interconnect network 1002 may include various metal layers/lines, vias, interlayer dielectrics, and/or other appropriate features. The multi-level interconnect network 1002 may, at various locations throughout the substrate, connect to the local interconnect provided by the metal layer 702, for example, by way of one or more VIAs passing through the fourth dielectric layer 902, and to provide electrical contact to the device 304 or other devices formed in the substrate 302. In general, the multi-level interconnect network 1002 may be configured to connect various devices (e.g., such as the device 304) or other features/devices to form a functional circuit that may include one or more planar MOSFETs and/or FinFET devices. In furtherance of the example, the multi-level interconnect network 1002 may include vertical interconnects, such as vias or contacts, and horizontal interconnects, such as metal lines. The various interconnection features may employ various conductive materials including copper, tungsten, and/or silicide. In at least one example, a damascene and/or dual damascene process is used to form a copper-containing multi-level interconnect network 1002.

While an example of the method 200 was described above as providing a local interconnect that electrically connects a device source and drain (e.g., the source 306 and the drain 308 of the device 304), it will be understood that this example is not meant to be limiting in any way. For instance, embodiments of the present disclosure may be implemented to provide local interconnects that are used, for example, to electrically connect a source, drain, and/or gate of a given device, or those of nearby devices, and/or to connect to other nearby active and/or passive devices and/or features. Additionally, embodiments of the present disclosure may be used to provide local interconnects which facilitate a vertical connection of one or more devices to an overlying metallization layer (e.g., through a dielectric layer and to the

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multi-level interconnect network 1002), for example, through one or more vias. As merely a few examples, a local interconnect as described herein may connect a source of a given device to a drain of a neighboring device, a gate of a given device to a source or drain of the given device, a source/drain/gate of a given device to a VIA that further connects the source/drain/gate of the given device to an overlying metal layer, or any of a plurality of other possible local interconnect connections. In addition, the various embodiments disclosed herein may be utilized, for example, in conjunction with any of a variety of device types such as a planar device (e.g., the transistor 100), a multi-gate device (e.g., the FinFET 150), a GAA device, an Ω -gate device, a Π -gate device, a strained-semiconductor device, an SOI device, a PD-SOI device, a FD-SOI device, or other device as known in the art. Moreover, additional process steps may be implemented before, during, and after the method 200, and some process steps described above may be replaced or eliminated in accordance with various embodiments of the method 200.

To elaborate further on the above discussion, reference is now made to FIGS. 11 and 12, which provide layout designs that illustrate various aspects of embodiments of the present disclosure. Referring first to FIG. 11, a layout design 1102 includes active region layers 1104, gate region layers 1106, local interconnect layers 1108, 1109, gate contact layers 1110, drain contact layers 1112, and a first layer of a multi-level interconnect network 1114. In some embodiments, the local interconnect layers 1108, 1109 represent local interconnects that may be formed as described above, and thus may for example include the metal layer 702. As shown in the example of FIG. 11, the local interconnect layers 1108, 1109 each extend over a single gate region layer 1106 within the active region layer 1104. Thus, the local interconnect layers 1108, 1109 may also be used to connect the source and drain of a given device, as described above. FIG. 11 also provides additional detail regarding local interconnect size (e.g., length and width). In one example, the local interconnect layers 1108, 1109 may be patterned to have a length 'L' equal to about 40-80 nanometers and a width 'W' equal to about 5-20 nanometers, similar to the slot size discussed above with reference to FIG. 5 and the slot 504. As noted above, the patterning of the slot will determine, at least in part, a size (e.g., length and width) of the subsequently formed local interconnect (e.g., the local interconnect layers 1108, 1109).

With reference to FIG. 12, a layout design 1202 is provided. The layout design 1202 is substantially similar to the layout design 1102; however, the layout design 1202 provides a local interconnect layer 1108A that may be used instead of the local interconnect layer 1108. In some embodiments, the local interconnect layer 1108A may be formed as described above, and thus may for example include the metal layer 702. As shown in the example of FIG. 12, the local interconnect layer 1108A extends over three gate regions 1106 within the active region 1104. In the example illustrated, the local interconnect layer 1108A may be used to connect the source and drain of a plurality of devices, or only the source and/or drain of selected devices over which the local interconnect layer 1108A spans, as well as for example the drain contact layer 1112. Generally, the local interconnect layers 1108, 1108A, or 1109, like the slot 504 discussed above with reference to FIG. 5, may be patterned in a variety of sizes, in accordance with a particular application, technology, or other process requirement. As merely one example, FIG. 12 illustrates an embodiment where the local interconnect layer 1108A is shown to be

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patterned to have a length equal to about $2.5\times$ the length of the local interconnect layer **1108**. Thus, in some examples, the local interconnect layer **1108A** may have a length 'L' equal to about 100-200 nanometers. Restating the above, in a broader sense, various embodiments of the local interconnects may be patterned to have a length 'L' equal to about 40-200 nanometers and a slot width 'W' equal to about 5-20 nanometers.

The various embodiments described herein offer several advantages over the existing art. It will be understood that not all advantages have been necessarily discussed herein, no particular advantage is required for all embodiments, and other embodiments may offer different advantages. As one example, embodiments discussed herein include methods and structures directed to a local interconnect fabrication process that provides a local interconnect without routing the local interconnect through an overlying metal layer. In particular, and in at least some embodiments, a local interconnect fabrication process is provided which provides the local interconnect routing through a VIA layer, without having to utilize interconnect layers of the multi-level network of metal wiring. As a result, and in various embodiments, the number of metal layers used in the multi-level interconnect network may be reduced (e.g., at least by one layer). Thus, the various embodiments disclosed herein provide for improved interconnect routing efficiency, a reduction in the required number of metal layers, cost reduction, and improved device and circuit performance.

Thus, one of the embodiments of the present disclosure described a method for fabricating a semiconductor device, where the method includes forming a first dielectric layer over a gate stack of at least one device and a second dielectric layer over a contact metal layer of the at least one device. In various embodiments, a selective etching process is performed to remove the second dielectric layer and expose the contact metal layer, without substantial removal of the first dielectric layer. In some examples, a metal VIA layer is deposited over the at least one device. The metal VIA layer contacts the contact metal layer and provides a local interconnect structure. In some embodiments, a multi-level interconnect network overlying the local interconnect structure is formed.

In another of the embodiments, discussed is a method where a device including a gate stack having a first dielectric layer formed thereon is provided. In some embodiments, a source and a drain are formed on either side of the gate stack. Additionally, and in various examples, a contact metal layer is disposed over the source and the drain, and a second dielectric layer is formed over the contact metal layer. By way of example, a third dielectric layer may be formed over the device and a first etching process of the third dielectric layer may be performed. In particular, the first etching process defines a slot within the third dielectric layer. Thereafter, in some embodiments, a second etching process is performed to remove the second dielectric layer and form openings that expose the contact metal layer over each of the source and the drain. In some examples, a conductive layer is deposited over the device, and within each of the slot and the openings. The conductive layer contacts the contact metal layer over each of the source and the drain.

In yet another of the embodiments, discussed is a semiconductor device including a substrate with a device having a gate stack and source/drain regions formed on either side of the gate stack. In some embodiments, a first dielectric layer is formed over the gate stack, and a contact metal is formed over the source/drain regions. By way of example, a third dielectric layer may also be formed over the device,

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where the third dielectric layer includes a slot defined therein. In some embodiments, a metal VIA layer is disposed over the device, where the metal VIA layer contacts the contact metal over both of the source/drain regions and provides a local interconnect between the source/drain regions. In addition, and in some embodiments, the metal VIA layer is separated from the gate stack by the first dielectric layer. In various examples, a multi-level interconnect network may be disposed over the metal VIA layer, where a fourth dielectric layer interposes the metal VIA layer and the multi-level interconnect network.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device, comprising:

a gate stack and source/drain regions formed on either side of the gate stack;

a first dielectric layer formed over the gate stack, and a contact metal formed over the source/drain regions, wherein the first dielectric layer includes a substantially uniform material composition extending between a first bottom surface of the first dielectric layer and a first top surface of the first dielectric layer, and wherein the first dielectric layer includes a nitride layer;

a local interconnect that contacts the contact metal over each of the source/drain regions to provide an electrical connection therebetween, wherein the local interconnect is separated from the gate stack by the first dielectric layer, wherein the local interconnect contacts the nitride layer of the first dielectric layer, wherein first portions of the local interconnect disposed over the source/drain regions have second bottom surfaces that are substantially level with the first bottom surface of the first dielectric layer, and wherein a second portion of the local interconnect disposed over the gate stack has a third bottom surface that is substantially level with the first top surface of the first dielectric layer.

2. The semiconductor device of claim 1, wherein a second top surface of the gate stack defines a first plane, and wherein a third top surface of the contact metal defines a second plane that is substantially level with the first plane.

3. The semiconductor device of claim 2, wherein lateral end portions of the local interconnect extend down to the second top surface of the contact metal on opposing sides of the first dielectric layer.

4. The semiconductor device of claim 1, further comprising:

a second dielectric layer disposed over the semiconductor device, wherein the second dielectric layer includes a slot defined therein, and wherein the local interconnect is disposed within the slot.

5. The semiconductor device of claim 4, wherein the first dielectric layer includes a first material and the second dielectric layer includes a second material different than the first material.

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6. The semiconductor device of claim 1, further comprising:

sidewall spacers disposed on sidewalls of the gate stack and the first dielectric layer.

7. The semiconductor device of claim 1, wherein the local interconnect includes Ti, W, Co, Cu, Al, Mo, MoW, TiN, TaN, WN, silicides, or combinations thereof.

8. The semiconductor device of claim 1, further comprising:

a multi-level interconnect network disposed over the local interconnect, wherein a third dielectric layer interposes the local interconnect and the multi-level interconnect network.

9. A semiconductor device, comprising:

a first dielectric layer formed over a gate stack and a contact metal layer formed over each of a source region and a drain region disposed on opposing sides of the gate stack, wherein the first dielectric layer includes a first oxide layer in contact with the gate stack; and

a metal layer that electrically connects the contact metal layer over the source region to the contact metal layer over the drain region, wherein a middle portion of the metal layer is electrically isolated from the gate stack by the first dielectric layer, wherein end portions of the metal layer contacting the source region and the drain region are laterally separated from the first dielectric layer by sidewall spacers disposed on sidewalls of the gate stack, and wherein the sidewall spacers include a second oxide layer that has a same composition as the first oxide layer;

wherein a first top surface of the gate stack defines a first plane, wherein a second top surface of the contact metal layer over the source region defines a second plane substantially coplanar with the first plane; and wherein a third top surface of the contact metal layer over the drain region defines a third plane substantially coplanar with both the first plane and the second plane.

10. The semiconductor device of claim 9, wherein the first dielectric layer has a first width, and wherein the gate stack has a second width that is substantially equal to the first width.

11. The semiconductor device of claim 9, further comprising:

a multi-level interconnect network disposed over the metal layer, wherein another dielectric layer interposes the metal layer and the multi-level interconnect network.

12. The semiconductor device of claim 9, wherein the sidewall spacers are disposed on the sidewalls of the gate stack and on sidewalls of the first dielectric layer.

13. The semiconductor device of claim 9, wherein the metal layer includes Ti, W, Co, Cu, Al, Mo, MoW, TiN, TaN, WN, silicides, or combinations thereof.

14. The semiconductor device of claim 9, further comprising:

a second dielectric layer formed over the semiconductor device, wherein the second dielectric layer includes a slot defined therein, and wherein the metal layer is disposed within the slot.

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15. The semiconductor device of claim 14, wherein the first dielectric includes a first material and the second dielectric includes a second material different than the first material.

16. The semiconductor device of claim 9, wherein the metal layer has a length equal to about 40-80 nanometers and a width equal to about 5-20 nanometers.

17. A semiconductor device, comprising:

a first gate stack and a second gate stack including source/drain regions disposed on either side of each of the first and second gate stacks, wherein the first gate stack has a first width and the second gate stack has a second width;

a first dielectric layer formed over the first gate stack and a second dielectric layer, separate from the first dielectric layer, formed over the second gate stack, wherein the first dielectric layer has a third width that is substantially equal to the first width, and wherein the second dielectric layer has a fourth width that is substantially equal to the second width; and

a local interconnect that electrically connects one of the source/drain regions disposed on either side of the first gate stack to one of the source/drain regions disposed on either side of the second gate stack, wherein the local interconnect is separated from the first gate stack by the first dielectric layer and from the second gate stack by the second dielectric layer;

wherein first portions of the local interconnect disposed over the one of the source/drain regions disposed on either side of the first gate stack and the one of the source/drain regions disposed on either side of the second gate stack have first bottom surfaces that are substantially level with second bottom surfaces of the first dielectric layer and the second dielectric layer, and wherein a second portion of the local interconnect disposed over the at least one of the first gate stack and the second gate stack has a third bottom surface that is substantially level with top surfaces of the first dielectric layer and the second dielectric layer.

18. The semiconductor device of claim 17, further comprising:

a first contact metal formed over and interposing the one of the source/drain regions disposed on either side of the first gate stack and a first one of the first portions of the local interconnect, and a second contact metal formed over and interposing the one of the source/drain regions disposed on either side of the second gate stack and a second one of the first portions of the local interconnect.

19. The semiconductor device of claim 17, wherein the first portions of the local interconnect are laterally separated from the first dielectric layer and the second dielectric layer by respective sidewall spacer layers.

20. The semiconductor device of claim 17, further comprising:

a third gate stack interposing the first and second gate stacks; and

a third dielectric layer, separate from the first and second dielectric layers, formed over the third gate stack; wherein the local interconnect is further separated from the third gate stack by the third dielectric layer.

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